

When Do Reproducible Scores Support a Valid Comparison in Dynamic Endoscopic Reconstruction?

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Abstract

In dynamic endoscopic reconstruction, performance tables guide baseline selection and challenge design. Prior work shows that evaluation choices change image-quality scores, while historical entries may use different frames, regions, targets, or averaging rules. Apparently comparable numbers can therefore describe different measurements. Using paper text, released source, and saved outputs, we ask: can the score be reproduced, can its saved outputs and exact source record be identified, and do two scores support the intended comparison? We applied this workflow to seven archived rows from our project. Five were reproduced; all 20 tested score cells matched the recorded values. Saved outputs from two methods had exact source bindings in both scenes, but their reported scores used different image regions. Rescoring those outputs on a shared dataset-defined region supported a direct comparison of the saved outputs; its margin and scene-level interpretation remained sensitive to the scoring choices. We then examined 27 results from eight papers in a predefined 16-paper surgical corpus. Across 162 checks of scored output, test cases, image region, representation, metric definition, and averaging, the public record contained useful components but insufficient detail to establish a common measurement. This study provides authors, reviewers, and challenge organisers with a practical path from reproducible scores to evidence that supports a fair comparison.

KEYWORDS

validation study, dynamic endoscopy, surgical scene reconstruction, benchmark comparability, Gaussian splatting

1 | INTRODUCTION

Dynamic endoscopic reconstruction aims to recover a changing surgical scene from video for research on visualisation, navigation, and scene understanding. Performance tables influence which techniques become baselines, which ideas are extended, and how challenges are designed. Those decisions are strongest when each score describes the same task and measurement as the score beside it.

Peak signal-to-noise ratio (PSNR) is a commonly reported image-similarity score, with higher values usually interpreted as better reconstruction fidelity. Two papers may both report “masked PSNR on EndoNeRF” while using different masks or crops, so different pixels enter the score. Each number may be correctly calculated, but their difference then mixes differences in the saved reconstructions with differences in what was scored. The *evaluation region* is the set of pixels included in

scoring; the evaluated frames, output target, image representation, metric definition, and averaging rule also determine what a score means.

Using paper text, released source, and saved outputs, we ask three practical questions. First, *reproduce*: can the available inputs and outputs regenerate the score? Second, *trace*: can its saved outputs and exact source record be identified? Third, *compare*: were both results measured on the same cases, output target, image region, representation, metric, and averaging rule for the conclusion being made? Together these questions identify the strongest comparison supported by the available evidence.

The paper makes three contributions. **First**, a three-step workflow identifies the strongest comparison supported by available evidence. **Second**, a worked validation of seven archived rows moves from exact score reproduction to a common-region comparison of two fixed output sets. **Third**, a source-located study of 27 reported results maps the information currently available and the evidence needed for fair

comparisons. The contribution is this consequence-bearing retrospective study: it applies established concerns about evaluation choices to determine which comparisons the surviving evidence supports.

2 | RELATED WORK

2.1 | Surgical reconstruction evaluation

The audited lineage begins with EndoNeRF¹. It includes neural-surface reconstruction². Gaussian-splatting methods then pursue real-time deformation^{3,4,5,6,7}, alternative representations or scale^{8,9,10,11,12}, and occlusion, sparse-view, or frequency-aware reconstruction^{13,14,15,16}. Results are commonly reported with PSNR, SSIM, and LPIPS on EndoNeRF, StereoMIS, or SCARED. Geometry-aware work argues that photometric metrics should be complemented by geometric evidence¹³. Our question is narrower: before interpreting even a photometric table, can its historical cells be shown to share a measurement and a source?

2.2 | Controlled and prospective protocol studies

NerfBaselines integrates implementations under standardized loaders and environments, reproduces many published NVS values, and shows that seemingly minor choices such as resizing, background, and metric implementation can move scores and, for 3DGS, orderings¹⁷. It standardizes evaluation forward. Mind the Gap uses matched-count spatial holdouts in static scenes and reports a 3–12 dB interpolation–extrapolation gap plus two seed-confirmed reversals¹⁸. It isolates one protocol axis experimentally. Surgical video instead holds out temporal neighbours of a deforming scene, so that specific spatial result does not transfer; its paired design remains a useful methodological precedent. Neither study asks what an incomplete historical production chain allows a reader to infer without rerunning the original experiment.

2.3 | Reporting, documentation, and interpretive layers

Metrics Reloaded and its companion catalogue connect validation metrics to the problem and document metric-related pitfalls^{19,20}; BIAS standardizes biomedical challenge reporting²¹, and CLAIM addresses clinical-AI manuscripts²². BenchmarkCards and BetterBench address benchmark documentation and lifecycle quality respectively^{23,24}. Evaluation Cards goes further than a static template: it composes benchmark,

evaluation-run, and model metadata, then computes four interpretive signals²⁵. Its provenance signal identifies the reporting party and inherited benchmark risks; its comparability signal flags sufficiently large divergence across variants or parties on a canonical metric path. The system surfaces warnings rather than enforcing pairwise claim admission, and its provenance signal is not strong result-to-run attribution.

These works remove broad framework novelty from our contribution. What remains is a consequence-bearing domain result: we retrospectively inspect paper text, released source, and frozen artifacts; the replay, attribution, and comparison decisions disagree on real surgical results; and applying them changes the interpretation of our own table.

3 | VALIDATING A REPORTED COMPARISON

3.1 | Three questions: reproduce, trace, and compare

A *result unit* is one reported scalar tied to a method label, dataset or scene support, output target, and source location. We make three decisions in order:

1. **Reproduce (replay).** Are the relevant saved-output bytes, exact evaluated units, and measurement definition available, and do they reproduce the archived value? Rows remain replayable, historical-only, blocked, or explicitly withdrawn; missing evidence is never manufactured.
2. **Trace (attribution).** Are saved outputs linked to an exact source record? Full run attribution also requires configuration and execution records; a paper repository alone supplies neither link.
3. **Compare.** In this claim-conditioned comparison, do both result units share the population, representation, target, support, metric semantics, and reduction required for the declared scalar, artifact-ordering, or method-capability claim? A repairable difference is labelled *rescore required*; evidence that cannot decide remains *unresolved*.

Practical rule. Compare like with like: use the same cases, output target, and image representation; score the same image region with the same metric and averaging rule; and keep each new score traceable to its saved outputs and producing run. If only the scoring region or averaging differs and the required outputs exist, rescore those outputs. If the required cases or outputs are missing, generate them before making a new comparison.

Eligibility is therefore pair- and claim-conditioned. The same result can be replayable, useful for artifact analysis, insufficiently attributed for a method claim, and inadmissible beside

a second native score. Content hashes bind the identity of evidence and measurement records; they do not decide scientific comparability. Our *runnable protocol* records exact evaluation fields, while the historical *reported protocol* records only what a publication and its released source support. We do not fill the latter from a reference implementation or from another paper in the lineage.

3.2 | Study 1: validating comparison evidence in our archived table

Before inspecting outcomes, we froze the 7-result/four-method/two-scene table previously used in our own denominator study: Deform3DGS on cutting; Endo-4DGS, EndoGaussian, and SurgicalGaussian on cutting and pulling. A row was replayable only if raw render/ground-truth/mask bytes, exact frame pairing, polarity, dimensions, and evaluator semantics were identified. For each admitted row we rebuilt four PSNR cells: full-array (“keep”) and selected-area (“true exclusion”) denominators, each under frame-mean and pooled-MSE reduction. The two source-bound methods available on both scenes were then assessed pairwise under their native masks and rescored on one dataset-defined common support: the EndoNeRF ground-truth tool mask, thresholded and inverted, with no method-specific spatial edit. A separately frozen sensitivity uses valid non-tool tissue, also removing dataset-invalid pixels. All comparisons describe only these saved, fixed outputs.

The denominator transport used as a supporting check follows directly from the selected-pixel count. If a mask selects N of HW pixels and excluded pixels contribute zero error, then

$$\begin{aligned} \text{MSE}_{\text{keep}} &= \frac{1}{HW} \sum_{i \in S} (\hat{x}_i - x_i)^2, \\ \text{MSE}_{\text{excl}} &= \frac{1}{N} \sum_{i \in S} (\hat{x}_i - x_i)^2, \end{aligned} \quad (1)$$

$$\text{PSNR}_{\text{keep}} - \text{PSNR}_{\text{excl}} = 10 \log_{10} \frac{HW}{N} = -10 \log_{10} \frac{N}{HW}.$$

This identity checks denominator handling; it does not establish comparison eligibility.

3.3 | Study 2: evidence available in the literature

The literature study uses a predefined, frozen, purposive, and non-exhaustive 16-paper scoping corpus assembled by citation chasing from EndoNeRF, sampled comparison tables, and targeted keyword search. Its purpose is source-located description rather than systematic-review coverage; no registered database query or inclusion-flow protocol was used. Eight papers contribute 27 predefined result units; three papers contribute 21

of them, and the other corpus papers provide source context. For every result unit we judge six measurement-definition dimensions—aggregation, metric semantics, population, representation, support, and target—plus strong result-to-run attribution, for $27 \times 7 = 189$ rows in total. Some paper-level evidence was shared across result units; the dependence counts are reported in the supplement. Each dimension closes only when every comparison-critical field it consumes is stated without unresolved shape or conflict. *Not located* records the limit of the frozen public evidence; a comparison that depends on that field remains unresolved until the evidence is supplied.

One model-assisted audit workflow completed the primary extraction and re-derived all 189 rows in three isolated 63-row contexts given answer-withheld instructions and frozen source text. The same workflow performed source-located resolution under author supervision, so differences between passes are reported as process diagnostics rather than inter-rater evidence. Provider/model-build identifiers and full execution transcripts were not preserved. Every state/closure disagreement and every unequal controlled value was source-adjudicated; an agreement-triggered QA lane remained available for defects on which both passes agreed. Frozen inputs, answers, ordering, and decisions are content-bound. Those receipts establish identity and sequence, while perfect isolation remains unverified.

4 | RESULTS

4.1 | Study 1: a common evaluation region enabled comparison of the replayed scores

The frozen inventory classified five rows as replayable and two Endo-4DGS rows as historical-only because their raw-byte provenance, exact native frame identifiers, and authored frame pairing were unavailable. No row was silently dropped or replaced. Across the five replayable rows, all 20 denominator-by-reduction cells regenerated the archived values at the recorded three-decimal precision. Numerical replay therefore succeeded.

Attribution and comparison returned different answers. Four replayable rows were source-bound; the Deform3DGS artifact had an explicit unbound source identity, so its exact parity supports only that frozen artifact. EndoGaussian and SurgicalGaussian were source-bound on both scenes, but their native mask bytes differed. Consequently, all eight per-scene native comparison cells required a rescore. The five values remain exact replay results; a cross-result conclusion first required rescoring on a shared region.

On the dataset-defined common support, all eight per-scene cells were admissible to comparison. With equal scene weighting, the SurgicalGaussian artifact exceeded the EndoGaussian

TABLE 1 EndoGaussian minus SurgicalGaussian PSNR for two frozen artifacts over cutting and pulling. Native rows describe different supports and admit no ordering. Common-support rows admit an artifact-level ordering only; negative values favour the SurgicalGaussian artifact. Equal scene weighting is used.

Support	Frame mean	Pooled MSE	Disposition
Native, method-specific	+0.070	−0.448	descriptive only; ordering withheld
Common dataset tool mask	−1.274	−1.273	SurgicalGaussian artifact higher
Valid non-tool tissue	−0.453	−0.944	aggregate only; scene directions differ

artifact by 1.274 dB under frame-mean reduction and 1.273 dB under pooled-MSE reduction (Table 1). These new scores use a benchmark-defined common region and support a direct comparison of the two fixed output sets. Method-level capability additionally requires matched training and production evidence, which was outside this comparison.

The valid-tissue sensitivity preserved the equal-scene aggregate direction but narrowed the frame-mean margin by about 64% and the pooled margin by about 26%. It also exposed scene heterogeneity: EndoGaussian’s artifact was higher on cutting by 0.605 dB (frame mean) and 0.297 dB (pooled), whereas SurgicalGaussian’s artifact was higher on pulling by 1.511 and 2.185 dB respectively. The supported interpretation is therefore the equal-scene aggregate for these outputs, with the scene-specific directions reported separately.

The denominator identity supplies an orthogonal measurement check. On the fixed Deform3DGS cutting artifact, frame-mean keep-denominator and true-exclusion PSNR were 38.770 and 37.343 dB, a 1.427 dB gap predicted by Eq. 1. The value is exactly replayable while the artifact’s unbound source prevents a method-level interpretation—the separation the audit is designed to preserve. Its role is to quantify how the denominator moves this fixed-output score.

4.2 | Study 2: the public record provides partial measurement definitions

The 27 result units came from eight result-bearing papers inside the frozen 16-paper corpus. After source adjudication, none of the 162 judgements, one per result unit and measurement dimension, was fully specified: 111 were partially specified, 50 were not located, and one contained conflicting population evidence (Table 2). The public record nevertheless provided useful components: metric family closed for 27/27 units, scored output target for 21/27, named scene support for 14/27, and whether stated evaluation rather than training frames entered the metric for 17/27. However, the actual rule selecting evaluation frames closed for 0/27. Strong inspectable result-to-run attribution also closed for 0/27; a general code release was not upgraded to scalar provenance. This strong rule was tightened

TABLE 2 Source-adjudicated measurement-definition judgements over 27 predefined result units. Every dimension has 27 judgements; a dimension is fully specified only when all of its required fields close.

Dimension	Not located	Partial	Conflicting	N/A	Fully specified
aggregation	16	11	0	0	0
metric_semantics	0	27	0	0	0
population	0	26	1	0	0
representation	24	3	0	0	0
support	10	17	0	0	0
target	0	27	0	0	0

after a non-answer-withheld pilot: the earlier loose paper-level rule closed 7/27. We report the 7/27 versus 0/27 sensitivity and keep attribution outside the 162 measurement-definition headline.

After source review, an answer-withheld re-derivation by the same auditor changed 71 of 189 final judgements. This consequence shows why a paper-level default or a single prose extraction was inadequate. Because both readings came from one workflow, their difference is reported as a process diagnostic; full agreement and ruling counts are reported in the supplement.

5 | DISCUSSION

5.1 | What the study enables

The self-audit shows how exact score reproduction can be developed into evidence for fair comparison. Five archived rows and all 20 tested score cells replayed exactly. For the two source-bound output sets, different native evaluation regions made common-region rescoring the next admissible step. That shared measurement then supported a direct comparison of the saved outputs. The result preserves the reproducible scores while aligning the comparative conclusion with the evidence available.

The literature study maps both the useful information already present and the details still needed for a common comparison. Metric family, output target, named scenes, and stated evaluation-frame support were often recoverable. A field

recorded as *not located* marks the current public evidence boundary; any comparison that depends on it remains pending until the evidence is supplied. This distinction matters in surgical reconstruction because performance tables guide baseline selection and challenge design.

5.2 | A practical path to fair, traceable comparisons

5.2.1 | Authors

For each headline result, report the dataset release and named scenes, exact evaluation frame list or generation rule, output target, pixel/crop support, mask bytes and polarity, metric implementation and data range, denominator, reduction, and scene weighting. Bind each scalar to the evaluator, configuration, artifact, and producing source. For public benchmarks, released code or an executable scoring artifact should be the normal expectation rather than an optional supplement.

5.2.2 | Reviewers and editors

Ask what comparison a table is meant to support—a scalar check, an artifact ordering, or a method capability claim—and require common measurement evidence at that strength. A reproduced cell supports its recorded measurement; a common-support rescore supports a new fixed-output analysis. Missing comparison-critical information lowers the admissible claim until it is supplied. Where neither public artifacts nor a controlled-access substitute exists, the result provides less verifiable and reusable comparison evidence.

5.2.3 | Challenge organisers

Publish versioned split files, masks, evaluator code or containers, reduction rules, and machine-readable receipts tying each submitted scalar to its run. Where patient privacy or licensing blocks public artifacts, provide an organiser-run evaluator, sealed container, or controlled-access escrow that preserves the same linkage for reviewers. Operational objects of this kind turn shorthand such as “7:1” or “masked PSNR” into a measurement that participants and reviewers can inspect.

The same measurement principle can transfer to medical-image segmentation challenges, general NVS, and generative-model benchmark tables: first establish what each number measures and where it came from, then admit the comparison supported by a shared measurement. These are candidate applications; the present data describe only the surgical scoping corpus.

5.3 | Clinical and translational relevance

Research decisions upstream of clinical translation should rely on comparisons that reflect the same task and measurement. This study supports that step for endoscopic visualisation, navigation, and scene-understanding research by auditing the evidence behind reported score comparisons. Patient benefit, device safety, and workflow impact remain separate targets for translational validation.

5.4 | Scope and limitations

The corpus is frozen, purposive, and non-exhaustive: 16 papers define the scoping context, while 27 result units from eight papers enter the formal matrix, with 21 units drawn from three papers. Its counts describe this sample rather than field-wide prevalence. The 189 rows reuse 73 determinations, including 117 paper-shared rows, so they are not 189 independent observations. Both passes and the source resolution belong to one model-assisted workflow; their difference is a process diagnostic, not inter-rater reliability. Answer withholding and receipts improve traceability but cannot prove zero leakage, and the missing model-build/transcript record further limits process reproducibility. Public sources may have changed after the freeze, and authors may possess private evidence absent from the audit.

The self-audit is also deliberately narrow. Two of seven archived rows are historical-only; the Deform3DGS replay has no bound exact source record; and the comparison uses two source-bound artifacts on two EndoNeRF scenes. Common-support and valid-tissue results describe those frozen artifacts under declared measurements. Population-level method ranking, capability, and clinical validity require additional evidence. PSNR denominator transport is exact under its assumptions, but it addresses only one component of measurement identity; frame selection, target, representation, and aggregation remain separate.

The reviewer package does not redistribute the raw medical-image/output arrays or third-party paper and code archives because their licences do not authorize an unrestricted anonymous copy. It can recompute aggregate arithmetic from the recorded outputs and verify content identities. Full re-scoring of the omitted arrays and independent source re-derivation require acquiring those inputs under their own terms.

6 | CONCLUSION

Dynamic endoscopic reconstruction benefits when reproducible scores are developed into evidence for fair comparison. In our archived table, five rows and all 20 tested score cells replayed exactly, while a shared evaluation region was required before

two saved output sets could be compared directly. The literature study then showed which measurement details are currently recoverable and which remain necessary for a common comparison. A practical pathway follows: reproduce the score, trace its saved outputs and source record, align the measurement, and state the conclusion supported at that level. This pathway can help authors, reviewers, and challenge organisers build more trustworthy comparison evidence for surgical reconstruction.

DATA AND CODE AVAILABILITY

A de-identified supplementary ZIP accompanies the double-blind submission. It contains the frozen audit records, source locators, replay outputs, comparison dispositions, selected inspection code, an exact EndoEval source snapshot, and a standard-library verifier. The same evaluator is available in the anonymous public peer-review repository <https://github.com/anon-endoeval/endoeval>, pinned to commit `1bbe9aeef55211e039c9c9b0dee7ccf3ecc890f7`; snapshot metadata in the ZIP binds that commit to its Git tree. The bundle was tested from a clean extraction. Its verifier checks package integrity, the evaluator snapshot identity, reported aggregate arithmetic from stored records, and formal literature-judgement counts. Raw medical-image and method-output arrays and copyrighted paper/code source archives are not redistributed; full re-scoring or source-negative verification requires obtaining those inputs under their own terms.

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